

Assignment 2 Report

I. Protocol Description and Optimization

1.1 Baseline MSI Architecture

The core system is a directory-based protocol designed for an asynchronous, unordered point-to-point network. To handle message reordering and potential race conditions, the design utilizes:

- Stable States: P_I (Invalid), P_S (Shared), and P_M (Modified).
- Transient States: Seven processor-side pending states (PI_S, PI_M, PS_M, PM_I, PM_IS, PM_IM, PS_I) to buffer incoming invalidations and track acknowledgments.
- Directory States: H_I, H_S, and H_M, supported by transient states H_MS and H_MM to manage ownership handoffs.

1.2 Speculative Read Optimization

The primary optimization is a speculative owner-directed read, which attempts to bypass the 3-hop directory latency (Requester → Home → Owner → Requester) in favor of a 2-hop completion.

- Mechanism: When a node receives an invalidation, it records the requester as a speculated_owner. A subsequent load in P_I issues a GetS_Spec directly to that node.
- Success Path: If the target is still in P_M, it provides data directly to the requester and sends a SpecOwnerToS message to the Home node to synchronize the directory metadata.
- Fallback Path: If the target has moved out of P_M, it returns a Spec_Fail, and the requester falls back to a standard GetS through the Home node.
- Deadlock Prevention: Speculative traffic is isolated on a dedicated virtual channel (VC3) to ensure it does not block critical protocol messages.

II. Verification

The optimization increases the state space due to the additional interleavings of speculative messages and fallback transitions.

Metric	Baseline Protocol	Optimized Protocol
Status	No error found	No error found
States Explored	1,574	7,877
Rules Fired	3,769	19,252

III. State Machine Analysis

The implemented state machine, as illustrated in the provided diagram, covers the complete lifecycle of both standard and speculative transactions.

3.1 Processor Controller Logic

- Entry Points: P_I initiates standard transactions (PI_S, PI_M) or speculative loads (PI_S_Spec).
- Convergence: Speculative transients (PI_S_Spec, PI_M_Spec, PS_M_Spec) resolve into stable states upon receiving Data or return to baseline paths upon Spec_Fail.
- Ownership Handling: P_M is responsible for servicing both forwarded requests from Home and direct speculative requests.

3.2 Home/Directory Controller Logic

- Metadata Repair: The transition $H_M \rightarrow H_S$ via the SpecOwnerToS message is critical; it ensures the directory correctly reflects that a speculative read has added a new sharer.
- Forwarding Flow: Home manages H_MS and H_MM states to track in-flight ownership changes until a FwdAck is received, confirming transaction completion.

This integrated model successfully balances aggressive latency reduction with the formal guarantees required for multiprocessor cache coherence.

IV. Diagram

